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Title of paper :
Small-Signal Stability and Dynamic Behaviors of a Hydropower Plant with an Upstream Surge Tank Using Different PID Parameters

1. What is the most important take home messages (conclusion) in the paper?

The most important take-home messages from the paper on small-signal stability and dynamic behaviors of a hydropower plant with an upstream surge tank are as follows:

1. An intermediate unstable region exists in the regulating regions (RRs) of the hydro turbine governing system (HTGS) with a surge tank. The system's stability is influenced by factors such as the parameters of the governor and water level fluctuations in the surge tank.

2. The relationship among the intermediate unstable regulating region, the critical surge tank area, and the classical Thoma stable area is crucial. Understanding these relationships helps in designing and optimizing hydropower plants effectively.

3. The water inertia in the penstock significantly impacts system stability. The critical surge tank area is found to be almost twice the Thoma area in the case study, indicating the limitations of traditional stability criteria, especially for low head hydropower plants with long penstocks.

4. Future research directions include further investigations into the interactions between the HTGS, water diversion system, and electric subsystems, as well as the development of a more general mathematical expression for the critical surge tank area for practical use.

These conclusions highlight the complexity of system stability in hydropower plants and emphasize the importance of considering various factors for optimal design and operation.

2. What is general motivation for the research?

The general motivation for the research on small-signal stability and dynamic behaviors of a hydropower plant with an upstream surge tank using different PID parameters can be summarized as follows:

1. Previous studies have provided insights into the stability of hydro turbine governing systems (HTGS) with surge tanks, but there are knowledge gaps regarding the interaction between oscillations in the HTGS and surge tanks. The research aims to address these gaps and provide valuable information for engineers designing and optimizing hydropower plants.

2. The traditional Thoma stable area criterion used for surge tank design neglects important factors such as water inertia in the penstock and the effect of the governor, making the stability analysis of the entire system more complex. The study seeks to enhance understanding of the stability characteristics of different regulating regions and the factors influencing system stability.

3. By investigating the intermediate unstable regulating region and its influencing factors, the research aims to improve the design and operation of hydropower plants. Understanding the relationship between the critical surge tank area, the intermediate unstable region, and the Thoma stable area is essential for ensuring system stability and performance.

Overall, the research is motivated by the need to advance knowledge in the field of hydropower plant stability analysis, optimize system design and operation, and address existing gaps in understanding the dynamic behaviors of HTGS with surge tanks.

3. What is the research gap from previous work (technical problem that the authors found in previous literatures)?

The research identified several gaps from previous work in the field of hydropower plant stability analysis. These gaps include:

1. Previous studies focused on simplified theoretical analyses that did not fully consider the effect of the governor on system stability. The numerical simulations primarily concentrated on optimizing control methods rather than comprehensively analyzing system dynamics.

2. The traditional Thoma stable area criterion used for surge tank design overlooked critical factors such as the water inertia of the penstock and the impact of the governor on system stability. This limitation made the stability analysis of the entire system more complex and less accurate.

3. The interaction between oscillations in the hydroturbine governing system (HTGS) and surge tanks was not thoroughly explored in previous literature. Understanding this interaction is crucial for determining the stability characteristics of different regulating regions and identifying factors that influence system stability.

4. The existence of an intermediate unstable regulating region in the regulating regions (RRs) of the HTGS with surge tanks was not previously mentioned. Investigating this region and its influencing factors is essential for improving the design and operation of hydropower plants.

By addressing these research gaps, the study aims to provide a more comprehensive understanding of system stability in hydropower plants and offer insights for optimizing system design and operation.

4. **What are solutions proposed by other researchers to solve the general research problem? What are disadvantages of these previous works?**

Previous researchers have proposed various solutions to address the general research problem of stability analysis and optimization of hydropower plants with surge tanks. Some of the solutions and their associated disadvantages are outlined below:

1. Analytical Criteria for Stability: Researchers like Jimenez and Chaudhry provided analytical criteria for the stability of isolated hydro units, considering the effects of water hammer. However, these criteria may not fully capture the complex dynamics of hydropower plants with surge tanks, leading to potential inaccuracies in stability assessments.

2. Phase Plane Method and Stability Criteria: Chaudhry et al. investigated the stability of closed surge tanks using the phase plane method and developed stability criteria. While this method offers insights into system stability, it may oversimplify the dynamics and overlook certain critical factors affecting stability, such as governor characteristics and water inertia.

3. Effect of Surge Tank Throttling: Studied the effect of surge tank throttling on system stability and concluded that throttle did not impact governor stability. However, this study may not have considered all relevant parameters influencing system stability, leading to potential oversights in the analysis.

4. Small Signal Analyses: Analyzed the stability of hydropower plants with upstream and downstream surge tanks using small signal analyses. While this approach provides valuable information on system stability, it may not fully capture the dynamic behaviors of the system under various operating conditions, potentially limiting the accuracy of stability assessments.

5. Mathematical Modeling and State-Space Equations: Researchers like Liang et al. introduced novel mathematical models of hydropower plants with surge tanks based on state-space equations to study dynamic behaviors. While these models offer insights into system dynamics, they may be limited in their ability to account for all influencing factors, leading to potential inaccuracies in stability predictions.

Overall, while previous works have contributed valuable insights into the stability analysis of hydropower plants with surge tanks, they may have limitations such as oversimplification of system dynamics, neglect of critical parameters, and potential inaccuracies in stability assessments. Addressing these limitations and incorporating a more comprehensive analysis of system dynamics could enhance the accuracy and effectiveness of stability assessments and optimization strategies for hydropower plants.

5. **In this paper, what is the proposed solution from the authors to solve the research gap and the general problem? According to the authors, why their proposed solution is important? Is there any improvement to previous works? How accurate the improvement?**

In the paper, the authors proposed a solution to address the research gap and the general problem of stability analysis and optimization of hydropower plants with surge tanks. The key aspects of their proposed solution and its significance are as follows:

1. Proposed Solution: The authors used the state space method to establish a stability analysis model of a hydroturbine governing system (HTGS) coupled with a water diversion system. They studied the stability characteristics of different regulating regions (RRs) based on the eigenvalues of the system and verified their findings through numerical simulations in the time domain. The authors also analyzed the dynamic system responses and dominating factors for governor parameters selected in different domains of the RRs.

2. Significance of Proposed Solution: The authors emphasized the importance of their proposed solution in uncovering the unstable regulating regions and understanding the interactions between oscillations in the HTGS and surge tanks. By identifying the intermediate unstable RR and investigating its influencing factors, the authors aimed to provide valuable insights for designing and optimizing hydropower plants. Understanding the relationship between the critical surge tank area, the intermediate unstable region, and the classical Thoma stable area is crucial for ensuring system stability and performance.

3. Improvement to Previous Works: The proposed solution builds upon previous research by addressing the gaps in understanding the stability characteristics of different RRs and the factors influencing system stability. By considering the interaction between oscillations in the HTGS and surge tanks, the authors offer a more comprehensive analysis of system dynamics compared to previous studies that may have oversimplified system behaviors.

4. Accuracy of Improvement: The improvement to previous works in this paper is significant in terms of identifying the intermediate unstable regulating region and investigating its influencing factors. By providing a more detailed analysis of system stability and dynamic behaviors, the authors' proposed solution enhances the understanding of critical factors affecting the stability of hydropower plants with surge tanks. The accuracy of this improvement is supported by the numerical simulations and theoretical analysis conducted in the study.

Overall, the authors' proposed solution offers a more comprehensive and detailed analysis of system stability in hydropower plants with surge tanks, addressing key research gaps and providing valuable insights for system design and optimization.

6. **How did the authors validate their proposed solution? Is the solution supported by valid data or experimental results? Is there any discrepancy between what authors claimed and what they wrote in the paper?**

The authors validated their proposed solution through a combination of theoretical analysis and numerical simulations in the paper. The validation process involved the following steps:

1. Theoretical Analysis: The authors used the state space method to establish a stability analysis model of the hydroturbine governing system (HTGS) coupled with a water diversion system. They analyzed the stability characteristics of different regulating regions (RRs) based on the eigenvalues of the system.

2. Numerical Simulations: The authors conducted numerical simulations in the time domain to verify their theoretical findings and study the dynamic system responses under different operating conditions. By comparing the simulation results with the theoretical analysis, the authors were able to validate the stability analysis model and the proposed solution.

3. Data and Experimental Results: The validation of the proposed solution was supported by the data and results obtained from the numerical simulations. The authors presented graphs, figures, and tables in the paper to illustrate the system responses, eigenvalues, and stability characteristics of HTGS with surge tanks.

4. Discrepancy: There is no apparent discrepancy between what the authors claimed and presented in the paper regarding the validation of their proposed solution. The theoretical analysis and numerical simulations were described in detail, and the results were consistent with the authors' claims regarding the stability analysis and optimization of hydropower plants with surge tanks.

Overall, the authors' proposed solution was supported by valid data and numerical results obtained from the theoretical analysis and simulations conducted in the study. The validation process demonstrated the effectiveness of the proposed solution in addressing the research gap and providing valuable insights for system design and optimization in hydropower plants.

7. **Try to analyze research gap, proposed solution, and how to evaluate the proposed solution. Do you find any problem from these three important parts? For example, in simulation-based research, do you think that the proposed solution can be implemented in real-time situation? Do you think that the proposed solution is robust and feasible for different setting/environment? How about the computational time?**

Analyzing the research gap, proposed solution, and evaluation of the proposed solution in the context of this paper:

1. Research Gap:
   - The research gap identified in the paper revolves around the need to understand the stability characteristics of different regulating regions in hydropower plants with surge tanks, particularly focusing on the interaction between oscillations in HTGS and surge tanks.
   - The gap lies in the lack of comprehensive analysis considering all influencing factors that affect system stability, especially the intermediate unstable regulating region, which has not been extensively studied before.

2. Proposed Solution:
   - The authors proposed a solution that involves using the state space method to establish a stability analysis model of the HTGS coupled with a water diversion system. They studied the stability characteristics of different regulating regions and verified their findings through numerical simulations in the time domain.
   - The proposed solution aims to provide insights into system dynamics, identify unstable regulating regions, and offer valuable information for designing and optimizing hydropower plants with surge tanks.

3. Evaluation of Proposed Solution:
   - The proposed solution was evaluated through theoretical analysis and numerical simulations, which demonstrated the effectiveness of the stability analysis model in capturing system dynamics and responses.
   - While the numerical simulations supported the proposed solution, there are potential challenges in implementing it in real-time situations. The computational time required for the simulations may be a limiting factor for real-time applications, especially in dynamic operational environments.
   - The robustness and feasibility of the proposed solution in different settings or environments may also be a concern, as the model's accuracy and performance could vary based on system parameters and conditions.

4. Potential Problems:
   - One potential problem could be the scalability and adaptability of the proposed solution to different hydropower plant configurations and operational scenarios. The model's robustness in handling diverse settings needs to be thoroughly evaluated.
   - Another issue could be the computational time required for the numerical simulations. If the solution is intended for real-time applications, optimizing the computational efficiency without compromising accuracy is crucial.

In summary, while the proposed solution addresses the research gap and provides valuable insights into system stability, there may be challenges related to real-time implementation, robustness in different environments, and computational efficiency that need to be carefully considered and addressed for practical applications.

8. What is scientific contribution in this paper? Did the authors solve old problem with new approach? Did the authors solve new problem with old approach? Did the authors solve new problem with new approach?

The scientific contribution of the paper on stability analysis and optimization of hydropower plants with surge tanks lies in several key aspects:
1. Solving Old Problem with New Approach:
   - The authors addressed the longstanding issue of understanding the stability characteristics of hydropower plants with surge tanks by introducing a new approach that considers the interaction between oscillations in the HTGS and surge tanks.

- By using the state space method and conducting numerical simulations, the authors provided a fresh perspective on system dynamics and regulating regions, offering a novel way to analyze and optimize the stability of hydropower plants.

2. Solving New Problem with Old Approach:
   - While the paper primarily focused on analyzing the stability of hydropower plants with surge tanks, it is possible that some aspects of the problem were addressed using established methods or approaches.
   - For example, the use of eigenvalue methods for stability analysis and theoretical modeling of the HTGS components may be considered traditional approaches in hydropower plant research.

3. Solving New Problem with New Approach:
   - The authors introduced a new problem by identifying the intermediate unstable regulating region in hydropower plants with surge tanks, which had not been extensively studied before.
   - They combined this new problem with a new approach involving state space modeling and numerical simulations to analyze system stability and dynamic behaviors, providing a fresh perspective on optimizing hydropower plant performance.

Overall, the scientific contribution of the paper lies in the authors' innovative approach to addressing stability issues in hydropower plants with surge tanks, combining theoretical analysis with numerical simulations to uncover the intermediate unstable regulating region and provide insights for system design and optimization. By tackling both old and new problems with a mix of traditional and novel approaches, the authors have contributed valuable knowledge to the field of hydropower plant research.

9. Try to read the discussion and conclusion. Do you find any suggestion for improvement in the discussion? What are disadvantages of the proposed solution?

In the discussion and conclusion of this paper, several suggestions for improvement and potential disadvantages of the proposed solution can be identified:

1. Suggestions for Improvement in the Discussion:
   - The authors could consider discussing the practical implications of their findings in more detail, such as how the insights gained from the stability analysis model and numerical simulations can be applied to real-world hydropower plant operations.
   - Providing recommendations for optimizing the governor tuning parameters or surge tank design based on the study's results could enhance the practical relevance of the research.

2. Disadvantages of the Proposed Solution:
   - Computational Complexity: One potential disadvantage of the proposed solution is the computational complexity associated with the state space

modeling and numerical simulations. High computational requirements may limit the real-time applicability of the solution in dynamic operational environments.
- Sensitivity to Parameters: The stability analysis model and optimization approach may be sensitive to variations in system parameters or operating conditions. In practice, uncertainties or changes in these parameters could affect the accuracy and reliability of the results.
- Implementation Challenges: Translating the theoretical findings and simulation results into practical implementation for hydropower plants may pose challenges. The complexity of the model and the need for accurate parameter estimation could hinder straightforward application in real-world settings.

By addressing these potential limitations and considering practical implications in the discussion section, the authors could provide a more comprehensive analysis of the proposed solution's strengths and weaknesses. Additionally, discussing avenues for future research to overcome these disadvantages and enhance the applicability of the solution could further enrich the discussion and conclusion of the paper.

10. **Do you find any part of this paper that you don't understand clearly? (e.g.: mathematical equation, algorithm of the proposed solution, validation steps, etc.). Try to find as many as possible parts that you don't understand. This list will guide you to study basic knowledge needed to better understand the paper**.

Based on the analysis of the paper on stability analysis and optimization of hydropower plants with surge tanks, here are some parts that may require further study to enhance understanding:

1. Mathematical Equations:
   - Equations related to the water inertia constant, head loss coefficients, and surge tank area calculations (e.g., Equations (1), (2), (6), (7), and (8)) may require a deeper understanding of fluid dynamics and hydropower plant components.

2. State Space Method:
   - The application of the state space method for establishing a stability analysis model may involve concepts from control theory and system dynamics that could benefit from further exploration.

3. Thoma Critical Surge Tank Area:
   - Understanding the Thoma critical surge tank area concept and its significance in hydropower plant stability analysis (mentioned in the text) may require familiarity with surge tank design criteria and hydraulics.

4. Sensitivity Analysis:
   - Conducting sensitivity analysis to investigate the influencing factors on the intermediate unstable region may involve statistical methods and sensitivity metrics that could be explored further.

5. System Parameters:
   - Parameters such as the promptitude time constant, permanent speed drop of the governor, and self-regulation coefficient of the unit (mentioned in the text) may require clarification on their roles in the hydropower plant system.

6. Validation Steps:
   - Understanding the validation steps for the stability analysis model through numerical simulations in the time domain may involve knowledge of simulation techniques, model validation procedures, and interpretation of simulation results.

7. Surge Tank Design:
   - Exploring the implications of reducing the cross-sectional area of the surge tank and head loss coefficients of the headrace tunnel on system stability may require insights into surge tank design principles and their impact on hydropower plant performance.

By delving deeper into these areas of study, such as fluid dynamics, control theory, hydraulics, and simulation methods, a better understanding of the paper's content and implications for hydropower plant stability analysis can be achieved.

RESEARCH COMPARATIVE TABLE BETWEEN 10 JOURNALS AND PROPOSED RESEARCH

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Title	Author	Year	Main Focus	Scope of Work	Key Findings	Methodologies	Key Practices	Head Level	Optimization Method
Small-Signal Stability and Dynamic Behaviors of a Hydropower Plant With an Upstream Surge Tank Using Different PID Parameters	G. Li, et al.	2021	Stability analysis and performance optimization of a hydroturbine governing system coupled with a water diversion system	Stability analysis, numerical simulations, performance optimization	Identification of stability characteristics, discovery of an intermediate unstable regulating region, and optimization suggestions	State space method, numerical simulations	State space method, stability analysis, numerical simulations	Not specified	Not specified
A review on remedial attempts to counteract the power generation compromise from draft tubes of hydropower plants	A. Muhirwa, et al.	2020	Remedial methods to mitigate detrimental off-load instabilities in draft tubes	Review of existing methods, categorization, assessment	Review of structural passive, fluidic active, and hybrid flow control methods to mitigate draft tube instabilities	Review of existing methods, categorized into structural passive, fluidic active, and hybrid flow control techniques	Structural passive methods, fluidic active techniques, hybrid flow control	Not specified	Not specified
Impact of the Minimum Head on Low-Head Hydropower Plants Energy Production and Profitability	B. Ceran, et al.	2020	Impact of minimum head on electricity production and economic viability of small hydropower plants	Mathematical modeling, simulation, economic analysis	Comparison of energy production and economic viability between Archimedes and Kaplan turbines at low-head locations	Mathematical modeling, simulation, economic analysis	Archimedes and Kaplan turbines, mathematical modeling, economic analysis	Low-head	Not specified
Influence of equipment size and installation height on electricity production in an Archimedes screw-based ultra-low head small hydropower plant and its economic feasibility	H. Lavric, et al.	2019	Simulation of electricity production in small hydropower plants with ultra-low heads	Simulation modeling, mathematical models	Simulation-based assessment of electricity production and operational benefits in ultra-low head small hydropower plants	Mathematical modeling, simulation	Simulation modeling, mathematical models	Ultra-low head	Not specified

Low-head hydropower as a reserve power source: A case study of Northeastern Illinois	Trevor l. Auth,et al.	2021	Integration of low-head hydropower with wind power to provide reserve power generation	Case study, estimation, modeling, analysis	Assessment of low-head hydropower as a reserve power source for wind power, limitations in combined operation, and economic considerations	Case study, estimation of hydropower potential, modeling and analysis	Wind power integration, low-head hydropower, reserve power generation	Low-head	Not specified
The Benefits of Variable Speed Operation in Hydropower Plants Driven by Francis Turbines	E. Bortoni, et al.	2019	Variable speed operation to improve efficiency in existing hydropower plants	Theoretical analysis, laboratory tests, simulation	Evaluation of efficiency recovery with variable speed operation in existing hydropower plants, with focus on the Furnas hydropower plant	Theoretical analysis, laboratory tests, simulation	Variable speed operation, digitalization	Not specified	Not specified
Experimental Study at the Reservoir Head of Run-of-River Hydropower Plants in Gravel Bed Rivers. Part I: Delta Formation at Operation Level	C. Sindelar, et al.	2020	Study of delta formation at the head of a run-of-river hydropower plant	Scaled physical model tests, experiments	Investigation of delta formation process at the head of a run-of-river hydropower plant and its impact on sediment management and flood risk	Scaled physical model tests, experiments	Physical model tests, sediment management strategies	Not specified	Not specified
Hydropower Case Study Collection: Innovative Low Head and Ecologically Improved Turbines, Hydropower in Existing Infrastructures, Hydropeaking Reduction, Digitalization and Governing Systems	E. Quaranta, et al.	2020	Showcase of innovative technologies and practices adopted in hydropower installations to minimize environmental impacts and improve sustainability	Case studies, analysis of innovations	Showcase of innovative technologies and practices adopted in hydropower installations to minimize environmental impacts and improve sustainability	Case studies, analysis of innovations in hydropower installations	Innovative hydropower technologies and practices	Not specified	Not specified

Design and implementation of an electromechanical control system for micro-hydropower plants	A. Ali, et al.	2020	Development of an electromechanical control system to maintain constant frequency in micro-hydropower plants	Simulation, hardware implementation, testing	Development and testing of an electromechanical control system to maintain constant frequency in micro-hydropower plants	Simulation in MATLAB Simulink, hardware implementation and testing	Electromechanical control system, constant frequency regulation	Micro-hydropower	Not specified
Energy production assessment in a complex hydropower development	A. Georgescu, et al.	2020	Evaluation of energy production and operation scenarios in a complex multi-reservoir hydropower development system	Numerical modeling, analysis	Evaluation of energy production and operation scenarios in a complex multi-reservoir hydropower development system	Numerical modeling in EPANET and GNU Octave	Multi-reservoir hydropower development, EPANET, GNU Octave	Not specified	Not specified
Technical Modelling of Pascal's Law Implementation for Multiple Tailrace in Horizontal Flow Hydroelectric Plant	**This (current) research**		**Development of electrical and mechanical system for hydroelectric plant in horizontal water flow to get a higher energi production.**	**Mathematical models and simulation modeling**	**Applying head at horizontal flow for hydroelectric plant to get a higher energy production.**	**Mathematical modeling and simulation**	**Inovative electrical and mechanical sub-system**	**Ultra-low to low head**	**Pascal's Law Implementation at Multiple Tailrace for Horizontal Flow**

The blue-inks are novelty of this research.